

A GRADED CONSTRUCTION OF THE FREE δ_p -RING OVER THE SPHERE

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ABSTRACT. We answer Question 3 of Carmeli–Nikolaus–Yuan on spherical lifts of free δ_p -rings. For every prime p , we construct a p -complete Moore $\mathbb{E}_\infty$ -ring R whose Tate–Frobenius δ_p -structure identifies $\pi_0 R$ with the p -completion of the free δ_p -ring on one generator. The construction is graded and proceeds weight by weight. Its main homotopical input is a duality calculation for the Mitchell–Priddy spectra $L(k)$,

$$[\Sigma^j L(k), \mathbb{S}_p] \cong \pi_j L(k) \oplus \pi_j L(k-1), \quad L(-1) = 0,$$

which implies the cohomotopy vanishing needed in the induction. We also construct an integral Moore $\mathbb{E}_\infty$ -ring with $\pi_0 \cong \mathbf{Z}\{x\}$ whose induced p -complete δ_p -structure is the free one.

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1. INTRODUCTION

Let p be a prime. Carmeli, Nikolaus and Yuan associate to every Moore $\mathbb{E}_\infty$ -ring a natural $\widehat{\delta}$ -structure on its degree-zero homotopy ring by means of the Tate-valued Frobenius [CNY26, Proposition 2.41]. This leads them to ask which $\widehat{\delta}$ -rings admit spherical lifts and, in particular, whether the free δ_p -ring on one generator lifts to the sphere [CNY26, Question 3]. Their question is a basic test case for the relation between power operations in commutative ring spectra and Frobenius lifts in ordinary algebra.

Write

$$\mathbf{Z}\{x\} = \mathbf{Z}[x_0, x_1, x_2, \dots], \quad \delta(x_i) = x_{i+1},$$

and let

$$A_p := \mathbf{Z}_p\{x\}_p^\wedge = \mathbf{Z}_p[x_0, x_1, x_2, \dots]_p^\wedge.$$

Equivalently, the associated Frobenius lift satisfies

$$\varphi(x_i) = x_i^p + px_{i+1}.$$

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1.1. Main result. Our main theorem gives an affirmative answer to the lifting problem.

Theorem 1.1 (Main theorem). *For every prime p there exists a p -complete Moore $\mathbb{E}_\infty$ -ring R_p together with an isomorphism of δ_p -rings*

$$\pi_0 R_p \cong A_p,$$

where the δ_p -structure on $\pi_0 R_p$ is the one induced by the Tate-valued Frobenius of [CNY26, Proposition 2.41].

Moreover there exists an integral Moore $\mathbb{E}_\infty$ -ring $R_{\mathbf{Z}}$ and a ring isomorphism

$$\pi_0 R_{\mathbf{Z}} \cong \mathbf{Z}\{x\}$$

such that, after p -completion, the induced δ_p -structure is the free one.

The p -complete lift admits a stronger graded form. There is a connected graded $\mathbb{E}_\infty$ -algebra

$$\mathcal{R} = \bigoplus_{w \geq 0} \mathcal{R}_w \in \text{CAlg}(\text{Gr}(\text{Sp}_p^\wedge))$$

with $\mathcal{R}_0 \simeq \mathbb{S}_p$, each $\mathcal{R}_w$ finite free over $\mathbb{S}_p$, and homogeneous classes

$$x_i \in \pi_0 \mathcal{R}_{p^i}, \quad i \geq 0,$$

such that

$$\mathbf{F}_p \otimes \mathcal{R} \simeq \mathbf{F}_p[x_0, x_1, \dots]$$

as graded $\mathbb{E}_\infty$ - $\mathbf{F}_p$ -algebras and

$$\delta(x_i) = x_{i+1}.$$

The underlying p -complete ring of $\mathcal{R}$ is R_p .

1.2. Idea of the construction. The grading isolates the obstruction theory. In the category of connected graded commutative algebras, the passage from weights $\leq n$ to weight $n + 1$ is controlled by a latching map

$$L_{n+1}(A) \longrightarrow A_{n+1}.$$

Its cofiber is a weightwise version of topological André–Quillen homology, denoted $\text{TAQ}_{n+1}(A)$. For the spherical monoid ring $\mathbb{S}[\mathbf{N}]$ these weightwise indecomposables are the stable symmetric-power layers. After p -completion, Nakaoka’s theorem implies that they vanish away from p -power weights; at weight p^k one obtains

$$\text{TAQ}_{p^k}(\mathbb{S}[\mathbf{N}])_p^\wedge \simeq \Sigma^k L(k),$$

where $L(k)$ is the Mitchell–Priddy spectrum.

Thus non- p -power weights are forced. At a p -power weight one has to attach one new class, and compatibility with the Tate–Frobenius determines the required attaching map on the reduced summand of $\Sigma_+^\infty B\Sigma_p$. The remaining obstruction groups reduce to

$$[\Sigma^{k-2} L(k), \mathbb{S}_p], \quad k \geq 2.$$

The second main ingredient of the paper is the vanishing of these groups. We prove the more precise formula

$$(1) \quad [\Sigma^j L(k), \mathbb{S}_p] \cong \pi_j L(k) \oplus \pi_j L(k-1).$$

The proof combines the equivariant Barratt–Priddy–Quillen theorem of Guillou–May [GM17], Segal completion in the form used by Ragnarsson [Rag11], Sankar’s

parabolic restriction formula for Steinberg modules [San19], and the Mitchell–Priddy splitting [MP83, AL10]. Connectivity of $L(k)$ then gives the required vanishing.

1.3. Relation with Carmeli–Nikolaus–Yuan. The only δ_p -structure used in the statement is the one supplied functorially by the Tate-valued Frobenius. In particular, after constructing the graded algebra we still have to identify its induced Frobenius lift with the free one. The key power-operation identity is

$$(2) \quad P_\Sigma(a) = a^p \varepsilon_\Sigma + \delta(a)(p\varepsilon_\Sigma - \mathrm{tr}_\Sigma),$$

for a degree-zero class a in a p -complete Moore $\mathbb{E}_\infty$ -ring. On the reduced summand $V \subset (\Sigma_+^\infty B\Sigma_p)_p^\wedge$ this becomes

$$P_\Sigma(a)|_V = -\delta(a)t_\Sigma.$$

The non-torsion class $t_\Sigma \in [V, \mathbb{S}_p]$ therefore detects $\delta(a)$. This is what converts the prescribed attachment at weight p^{i+1} into the identity $\delta(x_i) = x_{i+1}$.

1.4. Organization. Section 2 develops the weightwise extension formalism and computes the indecomposables of the spherical monoid ring. Section 3 records the Tate–Frobenius and power-operation input. Section 4 constructs the graded lift conditionally on a cohomotopy vanishing. Section 5 proves the duality formula (1) and hence the vanishing. Section 6 assembles the p -complete and integral lifts and proves Theorem 1.1.

1.5. Conventions. We write Sp for spectra and $\mathrm{Sp}_p^\wedge$ for p -complete spectra. The p -complete sphere is $\mathbb{S}_p$. For spectra X, Y we write $[X, Y] = \pi_0 \mathrm{Map}(X, Y)$. All tensor products in $\mathrm{Sp}_p^\wedge$ are p -completed. If X is graded, X_w denotes its weight- w part. We use the standard convention that $\delta(a) = (\varphi(a) - a^p)/p$ in a p -torsion-free δ_p -ring. References to *Higher Algebra* and *Higher Topos Theory* are abbreviated as [Lur17, Lur09].

2. GRADED COMMUTATIVE ALGEBRAS AND WEIGHTWISE TAQ

2.1. Truncations and latching objects. Let $\mathcal{C}$ be a stable presentably symmetric monoidal ∞ -category whose tensor product preserves colimits separately in each variable. Put

$$\mathrm{Gr}(\mathcal{C}) = \mathrm{Fun}(\mathbf{N}^{\mathrm{disc}}, \mathcal{C})$$

with Day convolution, so

$$(X \otimes Y)_w \simeq \bigoplus_{a+b=w} X_a \otimes Y_b.$$

Let $\mathrm{Gr}_{\leq n}(\mathcal{C})$ denote the full subcategory of objects supported in weights $\leq n$, and let $\tau_{\leq n}$ be weight truncation. A graded commutative algebra is *connected* if its weight-zero part is the tensor unit.

Lemma 2.1. *The localization $\tau_{\leq n}$ is compatible with the symmetric monoidal structure. Hence $\mathrm{Gr}_{\leq n}(\mathcal{C})$ is symmetric monoidal, and*

$$\tau_{\leq n} : \mathrm{CAlg}(\mathrm{Gr}(\mathcal{C})) \longrightarrow \mathrm{CAlg}(\mathrm{Gr}_{\leq n}(\mathcal{C}))$$

is left adjoint to the inclusion. The same holds between successive truncations.

Proof. For Y supported in weights $\leq n$ the projection $X \rightarrow \tau_{\leq n} X$ induces an equivalence $\text{Map}(\tau_{\leq n} X, Y) \simeq \text{Map}(X, Y)$. If f is an equivalence in weights $\leq n$, then so is $f \otimes Z$, since the weight- w piece for $w \leq n$ only involves weights at most n . The claim follows from the monoidal localization formalism of [Lur17, §2.2.1]. $\square$

Let

$$p_! : \text{CAlg}(\text{Gr}_{\leq n}(\mathcal{C})) \longrightarrow \text{CAlg}(\text{Gr}_{\leq n+1}(\mathcal{C}))$$

be the left adjoint to truncation. For $A \in \text{CAlg}(\text{Gr}_{\leq n}(\mathcal{C}))$ define the *latching object*

$$L_{n+1}(A) := (p_! A)_{n+1}.$$

If B is defined in weights at least through $n+1$, the counit gives a canonical map

$$\varepsilon_{B, n+1} : L_{n+1}(\tau_{\leq n} B) \longrightarrow B_{n+1}.$$

Proposition 2.2 (Weight-by-weight extension). *Let $A \in \text{CAlg}^c(\text{Gr}_{\leq n}(\mathcal{C}))$, put $L = L_{n+1}(A)$, and let $f : L \rightarrow M$ be a map in $\mathcal{C}$. Form*

$$K = \text{Free}_{\leq n+1}(L[n+1]), \quad Q = \text{Free}_{\leq n+1}(M[n+1])$$

and

$$B := p_! A \amalg_K Q.$$

Then $\tau_{\leq n} B \simeq A$, $B_{n+1} \simeq M$, and under these identifications the latching map of B is f . Moreover every connected algebra in weights $\leq n+1$ arises in this way, and for every B' there is a pullback square of mapping spaces

$$\text{Map}(B, B') \simeq \text{Map}(A, \tau_{\leq n} B') \times_{\text{Map}(L, B'_{n+1})} \text{Map}(M, B'_{n+1}).$$

Proof. Since $\text{Free}_{\leq n+1}(X[n+1])$ is supported in weights 0 and $n+1$, truncating the pushout to weights $\leq n$ gives A . In weight $n+1$, the relative tensor-product bar construction shows that no mixed positive-weight term can occur; hence the new weight is exactly M , and the latching map is f . Conversely, applying the same construction to the latching map of a given B' produces a map to B' which is an equivalence in every weight. The formula for mapping spaces follows by mapping out of the pushout and using the adjunction defining $p_!$. This is the standard latching-object description of graded commutative algebras, written explicitly in the present truncation convention. $\square$

2.2. Weightwise indecomposables.

Definition 2.3. For a connected graded commutative algebra B and $w \geq 1$, define

$$\text{TAQ}_w(B) := \text{cofib}(L_w(\tau_{< w} B) \xrightarrow{\varepsilon_{B, w}} B_w).$$

We write $\text{TAQ}(B) = \bigoplus_{w \geq 1} \text{TAQ}_w(B)[w]$.

The notation is justified by the following universal property.

Proposition 2.4. *Let B be connected.*

(i) *For $M \in \mathcal{C}$,*

$$\text{Map}_{\text{CAlg}}(B, \mathbf{1} \oplus M[w]) \simeq \text{Map}_{\mathcal{C}}(\text{TAQ}_w(B), M).$$

Consequently TAQ_w preserves colimits of connected algebras.

(ii) *If X is concentrated in positive weights, then $\text{TAQ}_w(\text{Free}(X)) \simeq X_w$.*

(iii) *$\text{TAQ}_w(B)$ depends only on $\tau_{\leq w} B$.*

- (iv) *Weightwise TAQ commutes with the symmetric monoidal base changes used below, in particular with p -completion and with $\mathbf{F}_p \otimes (-)$.*
- (v) *A map of connected graded algebras is an equivalence through weight n if it induces equivalences on TAQ_w for $w \leq n$.*
- (vi) *If r_d denotes rescaling of weights by d , then*

$$\mathrm{TAQ}_v(r_d B) \simeq \begin{cases} r_d \mathrm{TAQ}_{v/d}(B), & d \mid v, \\ 0, & d \nmid v. \end{cases}$$

Proof. For (i), truncate to weights $\leq w$ and apply [Theorem 2.2](#) to the square-zero target $\mathbf{1} \oplus M[w]$; the mapping space is the fiber over zero of

$$\mathrm{Map}(B_w, M) \longrightarrow \mathrm{Map}(L_w(\tau_{<w} B), M),$$

hence is $\mathrm{Map}(\mathrm{TAQ}_w(B), M)$. The remaining claims follow formally from this universal property, the free-forgetful adjunction, and the monoidal truncation of [Theorem 2.1](#). $\square$

2.3. The spherical monoid ring. Let

$$\mathbb{S}[\mathbf{N}] = \Sigma_+^\infty \mathbf{N}$$

with its grading by $\mathbf{N}$, so its generator t has weight 1. Iterated bar constructions identify its weightwise indecomposables with the stable symmetric-power filtration.

Theorem 2.5. *For every $w \geq 1$ there is a natural equivalence*

$$\mathrm{TAQ}_w(\mathbb{S}[\mathbf{N}]) \simeq \mathrm{colim}_k \Sigma^{-k} \Sigma^\infty (\mathrm{Sp}^w(S^k) / \mathrm{Sp}^{w-1}(S^k)).$$

After p -completion this spectrum vanishes unless $w = p^k$. For $w = p^k$ there is an equivalence

$$\mathrm{TAQ}_{p^k}(\mathbb{S}[\mathbf{N}])_p^\wedge \simeq \Sigma^k L(k),$$

where

$$L(k) := \Sigma^{-k} (\mathrm{Sp}^{p^k}(S^0) / \mathrm{Sp}^{p^k-1}(S^0))_p^\wedge, \quad L(0) = \mathbb{S}_p.$$

Proof. The bar construction of the free commutative monoid on a pointed set T is the free commutative monoid on $S^1 \wedge T$. Iterating gives

$$\mathrm{Bar}^k(\mathbb{S}[\mathbf{N}])_w \simeq \Sigma^\infty ((S^k)^{\wedge w} / \Sigma_w) \simeq \Sigma^\infty (\mathrm{Sp}^w(S^k) / \mathrm{Sp}^{w-1}(S^k)),$$

and the bar suspension maps agree with the classical suspension maps. Stabilizing yields the first assertion. Nakaoka's calculation of the mod- p cohomology of symmetric products implies that the p -completion of the layer is trivial unless the weight is a power of p ; see [\[Nak58, AD01\]](#). In weight p^k the remaining layer is the Mitchell–Priddy spectrum $\Sigma^k L(k)$, in the normalization of [\[MP83, AL10\]](#). $\square$

Set

$$c(k) = 2p^k - 2 - k.$$

Lemma 2.6. *For $k \geq 1$, the spectrum $L(k)$ is p -complete and connective, and*

$$\pi_j L(k) = 0 \quad (j < c(k)).$$

In particular $L(k)$ is 0-connected, $c(k) \geq k$, and $c(k-1) > k-2$ for $k \geq 2$.

Proof. Nakaoka's description of the mod- p cohomology of the symmetric-power layer shows that the first possible nonzero degree of $\Sigma^k L(k)$ is $2p^k - 2$; equivalently $H_j(L(k); \mathbf{F}_p) = 0$ for $j < c(k)$. The spectrum is bounded below and p -complete. The mod- p Hurewicz argument therefore upgrades the homology bound to the stated homotopy bound. The elementary inequalities follow immediately from the formula for $c(k)$; compare [Kuh15, §1] and [AL10, Proposition 4.11]. $\square$

Corollary 2.7 (Forced weights). *Let R be a connected graded $\mathbb{E}_\infty$ -algebra in p -complete spectra, with finite free components, and suppose*

$$\pi_0(\mathbf{F}_p \otimes R) \cong \mathbf{F}_p[x_0, x_1, \dots], \quad |x_i| = p^i.$$

Then $\mathrm{TAQ}_w(R) = 0$ when w is not a power of p . Thus in every non- p -power weight the latching map

$$L_w(\tau_{<w} R) \longrightarrow R_w$$

is an equivalence. At weight p^m the positive-degree part of the latching object is controlled by the spectra $\Sigma^{k-1} L(k)$, $1 \leq k \leq m$.

Proof. Over $\mathbf{F}_p$, a graded $\mathbb{E}_\infty$ -algebra with discrete components is determined by its ordinary graded commutative ring. Hence $\mathbf{F}_p \otimes R$ is the tensor product of weight-rescaled copies of $\mathbf{F}_p \otimes \mathbb{S}[\mathbf{N}]$. Apply Theorems 2.4 and 2.5. If w is not a p -power, the mod- p reduction of $\mathrm{TAQ}_w(R)$ vanishes; since the latter is connective and p -complete, it vanishes. $\square$

3. TATE-FROBENIUS AND GRADED POWER OPERATIONS

The construction in the previous section only sees the additive and multiplicative structure of a graded algebra. To recover the prescribed δ_p -structure we use the Tate-valued Frobenius of Nikolaus–Scholze [NS18] together with the comparison of Carmeli–Nikolaus–Yuan [CNY26, §2.4].

3.1. The induced graded δ_p -structure. Let $R = \bigoplus_w R_w$ be a connected graded $\mathbb{E}_\infty$ -algebra in p -complete spectra and suppose every R_w is a finite free $\mathbb{S}_p$ -module. Let

$$U := \bigoplus_w R_w \in \mathrm{CAlg}(\mathrm{Sp}), \quad R^u := U_p^\wedge.$$

Proposition 3.1. *The ring R^u is a p -complete Moore $\mathbb{E}_\infty$ -ring. Moreover:*

- (i) $\pi_0 U = \bigoplus_w \pi_0 R_w$ is a graded ring with finite free $\mathbf{Z}_p$ -components and

$$\pi_0 R^u \cong (\pi_0 U)_p^\wedge.$$

- (ii) *The Tate-valued Frobenius is homogeneous of degree p : there are maps*

$$\varphi_w : R_w \longrightarrow R_{pw}$$

whose images on π_0 satisfy

$$\varphi(\pi_0 R_w) \subseteq \pi_0 R_{pw}, \quad \delta(\pi_0 R_w) \subseteq \pi_0 R_{pw}.$$

Thus $\pi_0 U$ is a graded δ_p -ring.

- (iii) *The δ_p -structure on $\pi_0 R^u$ supplied by [CNY26, Proposition 2.41] is the unique p -adically continuous extension of this graded structure.*

Proof. The Moore assertion follows from the comparison

$$(X_p^\wedge) \otimes H\mathbf{Z} \simeq (X \otimes H\mathbf{Z})_p^\wedge$$

for bounded-below spectra [CNY26, Lemma 2.40]. Since $U \otimes H\mathbf{Z}$ is the discrete graded group $\bigoplus_w \pi_0 R_w$, this also gives the formula for $\pi_0 R^u$.

For homogeneity, write $\Delta : X \rightarrow (X^{\otimes p})^{tC_p}$ for the Tate diagonal and

$$\text{frob}_U : U \longrightarrow U^{tC_p}$$

for the Tate-valued Frobenius. Naturality of Δ and the graded multiplication show that the restriction to R_w factors through $R_{pw}^{tC_p}$. Since R_{pw} is finite free and p -complete, the trivial-action comparison

$$R_{pw} \xrightarrow{\sim} R_{pw}^{tC_p}$$

of [CNY26, Proposition 2.36] identifies this factor with a map $\varphi_w : R_w \rightarrow R_{pw}$. Hence both $\varphi(a)$ and a^p have weight pw , and p -torsion-freeness forces $\delta(a)$ to have the same weight. Naturality under $U \rightarrow U_p^\wedge$ proves the final assertion. $\square$

The following elementary recognition criterion will be used at the end of the construction.

Lemma 3.2 (Recognition of the free δ_p -ring). *Let*

$$B = \mathbf{Z}_p[y_0, y_1, \dots], \quad |y_i| = p^i,$$

be a graded p -torsion-free δ_p -ring. Write $I = B_{>0}$. Then

$$B_{p^{i+1}} = \mathbf{Z}_p y_{i+1} \oplus (I^2)_{p^{i+1}}$$

and there is a unique $c_i \in \mathbf{Z}_p$ with

$$\delta(y_i) \equiv c_i y_{i+1} \pmod{I^2}.$$

The following are equivalent:

- (i) *every c_i is a unit;*
- (ii) *the unique δ_p -map*

$$\mathbf{Z}_p\{x\} = \mathbf{Z}_p[x_0, x_1, \dots] \longrightarrow B, \quad x_0 \longmapsto y_0,$$

is an isomorphism;

- (iii) *B is isomorphic to the free δ_p -ring on one generator as a graded δ_p -ring.*

Proof. The usual identities for a p -derivation [BS22, Lemma 2.3 and Remark 2.2] imply $\delta(I^2) \subseteq I^2$. Hence, if $z \equiv vy_i \pmod{I^2}$, then

$$\delta(z) \equiv vc_i y_{i+1} \pmod{I^2}.$$

Inductively,

$$\delta^i(y_0) \equiv c_0 c_1 \cdots c_{i-1} y_i \pmod{I^2}.$$

If all c_i are units, the map $x_i \mapsto \delta^i(y_0)$ is triangular with unit diagonal and is therefore an isomorphism weight by weight. The converse follows by passing to indecomposables I/I^2 . $\square$

3.2. Transfer classes and the C_p -Segal input. For a finite G -set T , let

$$t_T \in \pi_0 \mathbb{S}^{hG}$$

be the class represented by the equivariant diagonal $\mathbb{S} \rightarrow \Sigma_+^\infty T$ followed by the fold map. If $T = G$ with the translation action, then t_G is the norm of the unit. We shall use

$$\mathrm{tr} = t_{C_p} \in \pi_0 \mathbb{S}^{hC_p}, \quad \mathrm{tr}_\Sigma = t_{\{1, \dots, p\}} \in \pi_0 \mathbb{S}^{h\Sigma_p}.$$

Let ε and ε_Σ denote the unit classes.

Proposition 3.3. *After p -completion one has*

$$[\Sigma_+^\infty BC_p, \mathbb{S}_p] \cong \mathbf{Z}_p \varepsilon \oplus \mathbf{Z}_p \mathrm{tr}, \quad \mathrm{tr}^2 = p \mathrm{tr}.$$

Restriction induces an injection

$$[\Sigma_+^\infty B\Sigma_p, X] \hookrightarrow [\Sigma_+^\infty BC_p, X]$$

for every p -complete spectrum X , and therefore

$$[\Sigma_+^\infty B\Sigma_p, \mathbb{S}_p] \cong \mathbf{Z}_p \varepsilon_\Sigma \oplus \mathbf{Z}_p \mathrm{tr}_\Sigma.$$

If

$$(\Sigma_+^\infty B\Sigma_p)_p^\wedge \simeq \mathbb{S}_p \oplus V$$

is the basepoint splitting, then

$$[V, \mathbb{S}_p] \cong \mathbf{Z}_p t_\Sigma, \quad t_\Sigma = (\mathrm{tr}_\Sigma - p\varepsilon_\Sigma)|_V,$$

and t_Σ is non-torsion.

Proof. The norm cofiber sequence and the Segal comparison

$$\mathbb{S}_p \xrightarrow{\sim} (\mathbb{S}_p)^{t_{C_p}}$$

give the first decomposition; this is exactly the form of the cyclic Segal input recorded in [CNY26, Proposition 2.36]. For the restriction statement, let $G = \Sigma_p$, $H = C_p$, and use the transfer associated to G/H . The composite restriction–transfer is multiplication by $[G : H] = (p-1)!$, hence is an equivalence mod p and therefore after mapping into a p -complete spectrum. The remaining statements follow by restricting ε_Σ and tr_Σ and using the basepoint splitting. $\square$

3.3. The power-operation formula.

Proposition 3.4. *Let R be a p -complete connective Moore $\mathbb{E}_\infty$ -ring with $\pi_0 R$ p -torsion-free. For $a \in \pi_0 R$, let*

$$P_\Sigma(a) \in [\Sigma_+^\infty B\Sigma_p, R]$$

be the total power operation and let $P(a)$ denote its restriction to C_p . If φ and δ are the Tate–Frobenius operations on $\pi_0 R$, then

$$\begin{aligned} P(a) &= a^p \varepsilon + \delta(a)(p\varepsilon - \mathrm{tr}) = \varphi(a)\varepsilon - \delta(a)\mathrm{tr}, \\ P_\Sigma(a) &= a^p \varepsilon_\Sigma + \delta(a)(p\varepsilon_\Sigma - \mathrm{tr}_\Sigma). \end{aligned}$$

Consequently

$$P_\Sigma(a)|_V = -\delta(a)t_\Sigma.$$

The same formula holds weightwise for a graded Moore algebra: if a has weight w , then $P_\Sigma(a)$ lands in weight pw and $\delta(a) \in \pi_0 R_{pw}$.

Proof. Nikolaus–Scholze identify the composite of the total power operation with the canonical map to Tate fixed points with the Tate-valued Frobenius [NS18, Corollary IV.1.10 and Lemma IV.1.20]. Thus

$$\text{can}(P(a)) = \text{triv}(\varphi(a)).$$

By Theorem 3.3, the kernel of can on π_0 is generated by tr . Hence

$$P(a) - \varphi(a)\varepsilon = b \text{tr}$$

for a unique $b \in \pi_0 R$. Restricting to the underlying nonequivariant class gives

$$a^p = \varphi(a) + pb,$$

so $b = -\delta(a)$. Injectivity of restriction from Σ_p to C_p gives the second formula. The graded statement follows by applying the formula to the underlying ring and projecting back to the weight- p summand, using Theorem 3.1. $\square$

We will also use the low-weight calculation for the spherical monoid ring. Let $\mathbb{S}_p[\mathbf{N}]$ denote the weightwise p -completion of $\mathbb{S}[\mathbf{N}]$.

Proposition 3.5. *Let $h : \text{Free}(\mathbb{S}_p[1]) \rightarrow \mathbb{S}_p[\mathbf{N}]$ classify the monoid generator t . Then $\tau_{<p}h$ is an equivalence, and in weight p the latching map is the projection*

$$(\Sigma_+^\infty B\Sigma_p)_p^\wedge \simeq \mathbb{S}_p \oplus V \longrightarrow \mathbb{S}_p.$$

Hence

$$\text{TAQ}_p(\mathbb{S}_p[\mathbf{N}]) \simeq \Sigma V, \quad \delta(t) = 0.$$

Proof. For $w < p$, the collapse $\Sigma_+^\infty B\Sigma_w \rightarrow \mathbb{S}$ is a mod- p equivalence because $p \nmid |\Sigma_w|$. In weight p , the power operation of the monoid generator is constant at pt , so it is the projection to the sphere summand. The final identity follows from Theorem 3.4 and the non-torsion of t_Σ . $\square$

4. CELL ATTACHMENTS AND THE INDUCTIVE CONSTRUCTION

Throughout this section all graded algebras are connected objects of $\text{CAlg}(\text{Gr}(\text{Sp}_p^\wedge))$.

4.1. Two attachment lemmas. The first lemma records the part of the module theory that is used repeatedly in the induction.

Lemma 4.1. *Let K be a connected graded commutative algebra with augmentation $K \rightarrow \mathbb{S}_p$. Every K -module M has a functorial filtration by weights whose n -th graded piece is, as a K -module,*

$$\text{gr}_n M \simeq \varepsilon^*(M_n[n]).$$

Consequently, for every right K -module P and $Z \in \text{Sp}_p^\wedge$,

$$P \otimes_K \varepsilon^*(Z[n]) \simeq (P \otimes_K \mathbb{S}_p) \otimes Z[n].$$

If P is supported in nonnegative weights and M in weights at least n , then

$$(P \otimes_K M)_v = 0 \quad (v < n), \quad (P \otimes_K M)_n \simeq P_0 \otimes M_n.$$

Proof. The full subcategory of modules supported in weights at least n is presentable and closed under colimits, hence its inclusion admits a right adjoint. On underlying graded spectra the resulting truncation is simply $\tau_{\geq n}$. A module concentrated in one weight is necessarily obtained by restricting scalars along the augmentation, because all positive-weight operations move it out of that weight. The relative tensor-product formula follows from the two-sided bar construction. $\square$

Lemma 4.2 (Attaching a single weight). *Let E be a connected graded commutative algebra, let $w \geq 1$, and let*

$$f : M \longrightarrow E_w, \quad g : M \longrightarrow N$$

be maps of p -complete spectra. Form

$$P := E \otimes_{\mathbb{S}_p\{M[w]\}} \mathbb{S}_p\{N[w]\}.$$

Then:

- (i) $E \rightarrow P$ is an equivalence in weights $< w$;
- (ii) in weight w there is a pushout square of spectra

$$\begin{array}{ccc} M & \xrightarrow{f} & E_w \\ g \downarrow & & \downarrow \\ N & \longrightarrow & P_w; \end{array}$$

(iii)

$$\mathrm{TAQ}(P) \simeq \mathrm{TAQ}(E) \amalg_{M[w]} N[w],$$

where $M[w] \rightarrow \mathrm{TAQ}(E)$ is the composite $M \xrightarrow{f} E_w \rightarrow \mathrm{TAQ}_w(E)$;

- (iv) *cell algebras remain cell algebras, with no new cells below weight w .*

Proof. Write $K = \mathbb{S}_p\{M[w]\}$ and $Q = \mathbb{S}_p\{N[w]\}$. The pushout is the relative tensor product $E \otimes_K Q$. In the two-sided bar construction, no positive-weight contribution from K or Q can occur in total weight $< w$, and in weight w the only new term is the displayed pushout. The TAQ statement follows from [Theorem 2.4](#), since weightwise TAQ preserves pushouts and sends a free algebra on $M[w]$ to $M[w]$. $\square$

The next observation lets us fill all weights between two consecutive powers of p without choices.

Lemma 4.3 (Non- p -power weights are forced). *Suppose B is defined in weights $< N$, has finite free components, and*

$$\pi_0(\mathbf{F}_p \otimes B) \cong \mathbf{F}_p[y_1, \dots, y_s]_{< N},$$

where every generator has p -power weight. If N is not a power of p , then the latching object $L_N(B)$ is finite free, its mod- p reduction is discrete with degree-zero part the weight- N polynomials in the y_i , and the extension obtained by taking the latching map to be the identity has $\mathrm{TAQ}_N = 0$.

Proof. By base change and [Theorem 2.5](#), the mod- p reduction of TAQ_N of the corresponding polynomial algebra is zero: every possible summand would require N/d_i to be a power of p , contradicting the hypothesis. Hence $\mathbf{F}_p \otimes L_N(B)$ is exactly the discrete weight- N polynomial piece. A p -complete Hurewicz argument shows that $L_N(B)$ is a finite free sphere module. Extending by the identity kills the weight- N indecomposable. $\square$

4.2. The basic prime-power blocks. Let $i_!^{(m)}$ denote the left adjoint from algebras supported in weights $\leq m$ to all graded algebras. Define

$$E^{(j)} := i_!^{(p^{j+1}-1)} \tau_{<p^{j+1}} \mathbb{S}_p[\mathbf{N}], \quad j \geq 0,$$

with counit

$$c_j : E^{(j)} \longrightarrow \mathbb{S}_p[\mathbf{N}].$$

Proposition 4.4. *The algebras $E^{(j)}$ have the following properties.*

- (i) c_j is an equivalence in weights $< p^{j+1}$, and $\mathrm{TAQ}_w(E^{(j)}) = 0$ for $w \geq p^{j+1}$.
- (ii) $E^{(0)} \simeq \mathbb{S}_p\{x\}$ is free on a class of weight 1. In weight p ,

$$E_p^{(0)} \simeq \mathbb{S}_p \oplus V,$$

and the counit is the projection to $\mathbb{S}_p$.

- (iii) If

$$\Lambda^{(j)} := \mathrm{fib}(E_{p^{j+1}}^{(j)} \rightarrow \mathbb{S}_p),$$

then

$$\Lambda^{(j)} \simeq \Sigma^j L(j+1), \quad E_{p^{j+1}}^{(j)} \simeq \mathbb{S}_p \oplus \Lambda^{(j)}.$$

- (iv) Killing the fiber in weight p^{j+1} advances the block:

$$E^{(j)} \otimes_{\mathbb{S}_p\{\Lambda^{(j)}[p^{j+1}]\}} \mathbb{S}_p \simeq E^{(j+1)}.$$

The same statements hold after rescaling weights by a power of p .

Proof. The first assertion is the adjunction defining $i_!$ together with the TAQ Whitehead criterion of [Theorem 2.4](#). The case $j = 0$ is [Theorem 3.5](#). The cofiber of

$$E_{p^{j+1}}^{(j)} \longrightarrow \mathbb{S}_p$$

is $\mathrm{TAQ}_{p^{j+1}}(\mathbb{S}_p[\mathbf{N}]) \simeq \Sigma^{j+1} L(j+1)$ by [Theorem 2.5](#), so its fiber is $\Sigma^j L(j+1)$. The extension splits because $L(j+1)$ is connective. Finally attach a cell killing this fiber and use the TAQ Whitehead criterion: the resulting algebra agrees with $\mathbb{S}_p[\mathbf{N}]$ through the next prime-power range and has no indecomposables above it, which identifies it with $E^{(j+1)}$. $\square$

For integers $n \geq 1$ and $0 \leq i \leq n-1$ set

$$E_i^{(n)} := r_{p^i} E^{(n-1-i)}, \quad \Lambda_i^{(n)} := \Sigma^{n-1-i} L(n-i).$$

Thus $E_i^{(n)}$ has a generator in weight p^i , agrees with the corresponding rescaled monoid ring below weight p^n , and

$$(E_i^{(n)})_{p^n} \simeq \mathbb{S}_p \oplus \Lambda_i^{(n)}.$$

In particular $\Lambda_{n-1}^{(n)} \simeq V$.

4.3. The conditional tower. The only stable-homotopy input needed by the inductive construction is the following vanishing.

Definition 4.5. We say that the prime p satisfies the *Mitchell–Priddy vanishing condition* if

$$(V_p) \quad [\Sigma^{k-2} L(k), \mathbb{S}_p] = 0 \quad (k \geq 2).$$

We prove in [Theorem 5.5](#) that (V_p) holds for every prime.

The auxiliary algebras $W(n, i)$ in the proof keep track of killing the excess prime-power layer in one generator at a time. We state the induction in a form that makes the data used later explicit.

Definition 4.6. For $n \geq 1$, *stage- n data* consist of connected graded algebras

$$W(n, -1) = \mathbb{S}_p, \quad W(n, 0), \dots, W(n, n-1),$$

maps

$$a_i : E_i^{(n)} \longrightarrow W(n, i), \quad 0 \leq i \leq n-1,$$

and equivalences

$$W(n, i) \otimes_{E_i^{(n)}} \mathbb{S}_p \simeq W(n, i-1).$$

Writing $R^{(n-1)} := W(n, n-1)$, we require:

(S1) $R^{(n-1)}$ is a cell algebra with cells in weights at most p^{n-1} ;

(S2) in weights $< p^n$, every component is finite free and

$$\pi_0(\mathbf{F}_p \otimes R^{(n-1)}) \cong \mathbf{F}_p[x_0, \dots, x_{n-1}]_{< p^n}, \quad |x_i| = p^i,$$

with $\delta(x_i) = x_{i+1}$ for $i < n-1$;

(S3) the map a_i sends the distinguished generator of $E_i^{(n)}$ to the image of x_i in $W(n, i)$.

Stage 1 is simply

$$W(1, 0) = \mathbb{S}_p\{x_0\}, \quad W(1, -1) = \mathbb{S}_p.$$

Theorem 4.7 (Induction theorem). *Assume (V_p) . From stage- n data one can construct stage- $(n+1)$ data. More precisely, if $R = R^{(n-1)}$, then there is a map*

$$K_n \longrightarrow \mathbb{S}_p$$

from the excess part of the weight- p^n latching object such that the pushout

$$(3) \quad R^{(n)} := R \otimes_{\mathbb{S}_p\{K_n[p^n]\}} \mathbb{S}_p\{x_n[p^n]\}$$

has the following properties:

(i) $R \rightarrow R^{(n)}$ is an equivalence in weights $< p^n$;

(ii) every $R_w^{(n)}$ with $w < p^{n+1}$ is finite free;

(iii)

$$\pi_0(\mathbf{F}_p \otimes R^{(n)})_{< p^{n+1}} \cong \mathbf{F}_p[x_0, \dots, x_n]_{< p^{n+1}};$$

(iv) for the induced graded δ_p -structure,

$$\delta(x_i) = x_{i+1} \quad (0 \leq i < n).$$

The construction is compatible with the auxiliary quotients $W(n, i)$ and therefore iterates indefinitely.

Proof. We give the four steps of the construction, including the point at which (V_p) is used.

Step 1: the weight- p^n excess. For $0 \leq i \leq n-1$, base change along

$$E_i^{(n)} \longrightarrow W(n, i)$$

and the augmentation of $E_i^{(n)}$ gives a cofiber sequence in weight p^n

$$X_i \longrightarrow W(n, i)_{p^n} \longrightarrow W(n, i-1)_{p^n}.$$

Using [Theorem 4.1](#) and the splitting

$$(E_i^{(n)})_{p^n} \simeq \mathbb{S}_p \oplus \Lambda_i^{(n)},$$

one obtains

$$X_i \simeq \mathbb{S}_p^{a_i} \oplus \Lambda_i^{(n)}.$$

All other filtration layers are finite free sphere modules.

Step 2: compatible projections. Inductively construct maps

$$\rho_i : W(n, i)_{p^n} \longrightarrow \mathbb{S}_p^{N_i}$$

which are isomorphisms on π_0 , and put $K(n, i) = \text{fib}(\rho_i)$. The fibers fit into cofiber sequences

$$(4) \quad \Lambda_i^{(n)} \longrightarrow K(n, i) \longrightarrow K(n, i-1), \quad K(n, -1) = 0.$$

Thus $K(n, i)$ has a finite filtration with layers

$$\Sigma^{k-1} L(k), \quad n-i \leq k \leq n.$$

To extend the projection at stage $i-1$ across $X_i \rightarrow W(n, i)_{p^n}$, the only obstruction lies in

$$[\Sigma^{-1} W(n, i-1)_{p^n}, \mathbb{S}_p^{a_i}].$$

After splitting off the free part, this is built from

$$[\Sigma^{-1} \Sigma^{k-1} L(k), \mathbb{S}_p] = [\Sigma^{k-2} L(k), \mathbb{S}_p],$$

which vanishes by (V_p) . The construction can be made at the level of arrows, so the cofiber sequences (4) and their null-homotopies are compatible with the quotient maps $W(n, i) \rightarrow W(n, i-1)$.

Step 3: adjoining x_n . Set $K_n = K(n, n-1)$. The first layer of (4) is

$$\Lambda_{n-1}^{(n)} \simeq V.$$

The restriction map

$$[K_n, \mathbb{S}_p] \longrightarrow [V, \mathbb{S}_p]$$

is surjective: the cokernel is controlled by the same groups in (V_p) . Choose

$$h : K_n \longrightarrow \mathbb{S}_p$$

whose restriction to V is $-t_\Sigma$. Form the pushout (3). By [Theorem 4.2](#), the old weights are unchanged and the weight- p^n piece becomes a free sphere module with one new degree-zero generator x_n . The latching calculation and [Theorem 4.3](#) then fill all weights

$$p^n < w < p^{n+1}$$

uniquely, giving the stated polynomial mod- p homotopy ring.

It remains to identify $\delta(x_{n-1})$. The class x_{n-1} is represented by the map from the free block $E_{n-1}^{(n)} = \mathbb{S}_p\{x_{n-1}\}$. In weight p^n , the restriction of its power operation to V is, by construction, the class $-x_n t_\Sigma$. On the other hand [Theorem 3.4](#) gives

$$P_\Sigma(x_{n-1})|_V = -\delta(x_{n-1})t_\Sigma.$$

Since multiplication by t_Σ is injective on the free weight- p^n summand, we obtain

$$\delta(x_{n-1}) = x_n.$$

Naturality of the Tate–Frobenius preserves the identities $\delta(x_i) = x_{i+1}$ for $i < n - 1$.

Step 4: propagation of the auxiliary quotients. For $0 \leq i < n$, define

$$W(n+1, i) := W(n, i) \otimes_{\mathbb{S}_p\{K(n, i)[p^n]\}} \mathbb{S}_p,$$

and put $W(n+1, n) = R^{(n)}$. The cofiber sequences (4), together with [Theorem 4.4](#), identify the relevant base changes of the $E_i^{(n)}$ with $E_i^{(n+1)}$. The arrow-level compatibility from Step 2 ensures that the two possible null-homotopies agree, so

$$W(n+1, i) \otimes_{E_i^{(n+1)}} \mathbb{S}_p \simeq W(n+1, i-1).$$

The generator condition is inherited from stage n , and the properties proved in Step 3 are exactly the remaining stage- $(n+1)$ conditions. $\square$

Remark 4.8. No parity assumption on p enters [Theorem 4.7](#). At $p = 2$ certain adjacent cohomotopy groups such as $[\Sigma L(1), \mathbb{S}_2]$ need not vanish, so it is important that the construction uses coherent maps of cofiber sequences rather than arbitrary choices of null-homotopies. The only vanishing required is (V_p) itself.

5. DUALITY FOR THE MITCHELL–PRIDDY SPECTRA

We now prove (V_p) . The argument first computes the Spanier–Whitehead dual of the Steinberg summand of the suspension spectrum of an elementary abelian p -group and then passes to the Mitchell–Priddy summands.

5.1. Steinberg idempotents.

$$V_k = (\mathbf{F}_p)^k, \quad G_k = \mathrm{GL}_k(\mathbf{F}_p), \quad Y_k = (\Sigma_+^\infty BV_k)_p^\wedge.$$

The group G_k acts on Y_k by functoriality. Let $\Sigma_k \subset G_k$ be the permutation matrices and $B_k \subset G_k$ the upper triangular subgroup. In $\mathbf{Z}_{(p)}[G_k]$ set

$$\bar{\Sigma}_k = \sum_{\sigma \in \Sigma_k} \mathrm{sgn}(\sigma) \sigma, \quad \bar{B}_k = \sum_{b \in B_k} b, \quad c_k = \prod_{i=1}^k (p^i - 1),$$

and define

$$e_k = c_k^{-1} \bar{\Sigma}_k \bar{B}_k, \quad \hat{e}_k = c_k^{-1} \bar{B}_k \bar{\Sigma}_k.$$

The Steinberg relation implies that these are idempotents; see [\[San19, §2.1\]](#). Write

$$\mathrm{St}_k = \mathbf{Z}_p[G_k] e_k, \quad M(k) = e_k Y_k, \quad M(0) = \mathbb{S}_p.$$

We use two elementary consequences. First, for every G_k -module N ,

$$e_k N \cong \mathrm{Hom}_{G_k}(\mathrm{St}_k, N), \quad e_k N \cong \hat{e}_k N$$

as $\mathbf{Z}_p$ -modules. Second, if $1 \leq r \leq k-1$, $W \subset V_k$ has dimension r , and P_W is its stabilizer, then Sankar’s parabolic restriction formula gives

$$(5) \quad \mathrm{Res}_{P_W}^{G_k} \mathrm{St}_k \cong \mathrm{Ind}_{\mathrm{GL}(W) \times \mathrm{GL}(W')}^{P_W} (\mathrm{St}_W \otimes \mathrm{St}_{W'}),$$

for any complement W' of W ; see [\[San19, Propositions 8 and 10\]](#).

5.2. A transfer model for the dual of BV . For a finite group V , let $\mathcal{F}_V$ be the symmetric monoidal groupoid of finite left V -sets and equivariant bijections, with disjoint union. Let $K\mathcal{F}_V$ be its additive group-completion spectrum. Evaluation of a V -set along BV gives a canonical map

$$\Delta_V : K\mathcal{F}_V \longrightarrow F(\Sigma_+^\infty BV, \mathbb{S}).$$

Proposition 5.1. *If V is a finite p -group, then after p -completion the map Δ_V is an equivalence*

$$(K\mathcal{F}_V)_p^\wedge \xrightarrow{\sim} F(\Sigma_+^\infty BV, \mathbb{S}_p).$$

Under the equivariant Barratt–Priddy–Quillen equivalence of Guillou–May [GM17, Theorems 5.9 and 6.2], this is the canonical Segal-completion map for the genuine V -sphere.

Proof. Guillou–May identify $K\mathcal{F}_V$ with the V -fixed points of the genuine V -sphere and identify the evaluation map with the canonical map from genuine fixed points to Borel fixed points. The Segal conjecture identifies the latter with derived completion at the augmentation ideal of the Burnside ring. For a finite p -group, Ragnarsson’s comparison of augmentation-ideal completion with p -completion on the nontrivial-isotropy quotient [Rag11, Proposition 6.4] shows that, after p -completion, the canonical map is an equivalence. Since genuine fixed points preserve the relevant limits, this gives the displayed equivalence. The construction is natural in automorphisms of V because both the equivariant BPQ map and the finite-cover transfer are natural. $\square$

Decompose a finite V -set according to its stabilizer. Since V is abelian, every orbit is V/H for a unique subgroup $H \leq V$, and ordinary BPQ with labels gives a splitting

$$K\mathcal{F}_V \simeq \bigvee_{H \leq V} \Sigma_+^\infty B(V/H).$$

The transfer model makes this splitting canonical enough to retain the automorphism action.

Corollary 5.2 (Automorphism-equivariant dual splitting). *Let $V = V_k$ and $G = G_k$. There is a G -equivariant equivalence*

$$(6) \quad \bigvee_{H \leq V} (\Sigma_+^\infty B(V/H))_p^\wedge \xrightarrow{\sim} F(\Sigma_+^\infty BV, \mathbb{S}_p),$$

where $g \in G$ carries the H -summand to the gH -summand by the quotient isomorphism $V/H \xrightarrow{\sim} V/gH$, and acts on the target by precomposition with Bg^{-1} .

Proof. For $H \leq V$, let

$$\Gamma_H = \{(v, w) \in V \times V/H : q_H(v) = w\}.$$

The H -summand is the adjoint of the finite-cover transfer

$$\Sigma_+^\infty (BV \times B(V/H)) \longrightarrow \Sigma_+^\infty B\Gamma_H \longrightarrow \mathbb{S}_p.$$

These summands are exactly the orbit components of the evaluation map in [Theorem 5.1](#). The isomorphism

$$(v, w) \longmapsto (gv, \bar{g}_H w)$$

identifies the cover for H with the cover for gH , and these identifications compose strictly. Functoriality of finite-cover transfer therefore supplies the coherent G -action in (6). $\square$

5.3. The Steinberg summand. Let

$$D(X) = F(X, \mathbb{S}_p).$$

Proposition 5.3. *For every $k \geq 1$ and every $j \in \mathbf{Z}$ there is an isomorphism of finitely generated $\mathbf{Z}_p$ -modules*

$$\pi_j D(M(k)) \cong \pi_j M(k) \oplus \pi_j M(k-1).$$

Proof. Let

$$N_j = [\Sigma^j Y_k, \mathbb{S}_p]$$

with the left G_k -action $g \cdot f = f \circ \Sigma^j B g^{-1}$. If ι denotes the anti-involution of the group algebra induced by $g \mapsto g^{-1}$, then the image of the Steinberg idempotent on the source becomes $\iota(e_k) = \widehat{e}_k$ on N_j . Hence

$$\pi_j D(M(k)) \cong \widehat{e}_k N_j \cong e_k N_j.$$

By Theorem 5.2,

$$N_j \cong \bigoplus_{H \leq V_k} \pi_j(\Sigma_+^\infty B(V_k/H))_p^\wedge$$

as a G_k -module. Group the summands according to $r = \dim H$. Fix an r -dimensional subspace W_r and let P_r be its stabilizer. The corresponding part is induced from

$$X_{W_r} := \pi_j(\Sigma_+^\infty B(V_k/W_r))_p^\wedge.$$

Therefore Frobenius reciprocity gives

$$e_k \operatorname{Ind}_{P_r}^{G_k} X_{W_r} \cong \operatorname{Hom}_{P_r}(\operatorname{Res}_{P_r}^{G_k} \operatorname{St}_k, X_{W_r}).$$

For $r = 0$ this is $\pi_j M(k)$. For $r = k$ it is zero when $k \geq 2$, while for $k = 1$ it gives $\pi_j M(0)$. For $1 \leq r \leq k-1$, use (5). The factor $\operatorname{GL}(W_r)$ acts trivially on X_{W_r} , so the coinvariants of St_{W_r} vanish unless $r = 1$. Thus all intermediate dimensions except $r = 1$ disappear. When $r = 1$, the surviving factor is precisely the Steinberg summand for $V_k/W_r \cong V_{k-1}$, namely $\pi_j M(k-1)$. Summing the two surviving pieces proves the formula. $\square$

Mitchell and Priddy identify the spectrum $M(k)$ with two consecutive symmetric-power layers.

Theorem 5.4 (Duality formula). *For every prime p , every $k \geq 0$, and every $j \in \mathbf{Z}$,*

$$[\Sigma^j L(k), \mathbb{S}_p] \cong \pi_j L(k) \oplus \pi_j L(k-1), \quad L(-1) = 0.$$

Proof. By the Mitchell–Priddy splitting [MP83] in the formulation of [AL10, Propositions 2.9–2.10],

$$M(k) \simeq L(k) \vee L(k-1),$$

and $M(k)$ is the Steinberg summand of Y_k . After p -completion this agrees with the normalization of $L(k)$ in Theorem 2.5. Put

$$a_k = \pi_j D(L(k)), \quad b_k = \pi_j L(k), \quad b_{-1} = 0.$$

The spectra involved are retracts of suspension spectra of finite type, so these are finitely generated $\mathbf{Z}_p$ -modules. Dualizing the splitting and applying [Theorem 5.3](#) gives

$$a_k \oplus a_{k-1} \cong (b_k \oplus b_{k-1}) \oplus (b_{k-1} \oplus b_{k-2}).$$

For $k = 0$, $a_0 = b_0$. Induction and cancellation for finitely generated modules over the PID $\mathbf{Z}_p$ give

$$a_k \cong b_k \oplus b_{k-1}.$$

□

Corollary 5.5. *The Mitchell–Priddy vanishing condition (V_p) holds for every prime p :*

$$[\Sigma^{k-2}L(k), \mathbb{S}_p] = 0 \quad (k \geq 2).$$

More generally $[\Sigma^j L(k), \mathbb{S}_p] = 0$ for $j < 0$.

Proof. By [Theorem 5.4](#),

$$[\Sigma^{k-2}L(k), \mathbb{S}_p] \cong \pi_{k-2}L(k) \oplus \pi_{k-2}L(k-1).$$

Both terms vanish by [Theorem 2.6](#): one has $k-2 < c(k)$ and $k-2 < c(k-1)$. The statement for negative j follows from connectivity. □

6. ASSEMBLY AND SPHERICAL LIFTS

We now assemble the finite-weight stages, pass to the underlying ungraded algebra, and finally globalize the construction from the p -complete sphere to the sphere. Throughout this section the prime p is fixed.

Theorem 6.1 (The graded lift). *There is a connected graded $\mathbb{E}_\infty$ -algebra*

$$\mathcal{R} \in \mathrm{CAlg}^c(\mathrm{Gr}(\mathrm{Sp}_p^\wedge))$$

with $\mathcal{R}_0 \simeq \mathbb{S}_p$ and every $\mathcal{R}_w$ a finite free $\mathbb{S}_p$ -module, together with classes

$$x_i \in \pi_0 \mathcal{R}_{p^i}, \quad i \geq 0,$$

such that

$$\mathbf{F}_p \otimes \mathcal{R} \simeq \mathbf{F}_p[x_0, x_1, \dots]$$

as graded $\mathbb{E}_\infty$ - $\mathbf{F}_p$ -algebras, where $|x_i| = p^i$, and

$$\pi_0 \left(\bigoplus_{w \geq 0} \mathcal{R}_w \right) \cong \mathbf{Z}_p[x_0, x_1, \dots].$$

For the δ_p -structure induced by the Tate-valued Frobenius one has

$$\delta_p(x_i) = x_{i+1} \quad (i \geq 0).$$

Proof. By [Theorem 5.5](#), the vanishing hypothesis required in [Theorem 4.7](#) holds. Starting from the initial stage and applying the induction theorem repeatedly gives compatible stage- n data for every $n \geq 1$.

Let $R^{(n)}$ denote the algebra produced at stage n , and put

$$T_n := \tau_{< p^{n+1}} R^{(n)}.$$

The stable-range assertion in [Theorem 4.7](#) identifies

$$\tau_{< p^n} T_n \simeq T_{n-1}$$

compatibly with the classes $x_0, \dots, x_{n-1}$. For each $m \geq 0$, choose any n with $p^{n+1} > m$ and set

$$\mathcal{R}[m] := \tau_{\leq m} T_n.$$

The stable-range equivalences show that this is independent of the choice of n , up to a canonical compatible equivalence, and that

$$\tau_{\leq m} \mathcal{R}[m+1] \simeq \mathcal{R}[m].$$

The inverse limit

$$\mathcal{R} := \lim_m \mathcal{R}[m]$$

therefore exists in graded commutative algebras and satisfies

$$\tau_{\leq m} \mathcal{R} \simeq \mathcal{R}[m].$$

In each fixed weight all assertions are already true at a sufficiently large finite stage. It follows that every $\mathcal{R}_w$ is finite free over $\mathbb{S}_p$ and that

$$\mathbf{F}_p \otimes \mathcal{R} \simeq \mathbf{F}_p[x_0, x_1, \dots]$$

weightwise. Since both sides in each weight are finite free over $\mathbf{Z}_p$ and the comparison is an isomorphism modulo p , Nakayama's lemma gives

$$\pi_0 \left(\bigoplus_w \mathcal{R}_w \right) \cong \mathbf{Z}_p[x_0, x_1, \dots].$$

Finally, the Tate–Frobenius δ_p -operation is natural and homogeneous by [CNY26, Proposition 2.41]. For fixed i , choose a stage past weight p^{i+1} . At that finite stage the induction imposes $\delta_p(x_i) = x_{i+1}$, hence the same identity holds in $\mathcal{R}$. $\square$

Theorem 6.2 (The p -complete spherical lift). *Let*

$$A := \widehat{\mathbf{Z}_p\{x\}_{\delta_p}} = \widehat{\mathbf{Z}_p[x_0, x_1, \dots]_p}, \quad \delta_p(x_i) = x_{i+1}.$$

There is a p -complete Moore $\mathbb{E}_\infty$ -ring R_p and an isomorphism of δ_p -rings

$$\pi_0 R_p \xrightarrow{\sim} A,$$

where the δ_p -structure on $\pi_0 R_p$ is the one induced by the Tate-valued Frobenius. The underlying spectrum of R_p can be chosen of the form

$$\left(\bigvee_I \mathbb{S} \right)_p^\wedge$$

for a countable set I .

Proof. Let

$$U := \bigoplus_{w \geq 0} \mathcal{R}_w$$

be the underlying ungraded $\mathbb{E}_\infty$ -ring of the graded algebra of Theorem 6.1, and set $R_p := U_p^\wedge$. The monomials in the variables x_i form a homogeneous $\mathbf{Z}_p$ -basis of $\pi_0 U$. Since every graded piece is a finite free $\mathbb{S}_p$ -module, the underlying spectrum of U is a wedge of copies of $\mathbb{S}_p$, indexed by those monomials; hence the asserted description of the underlying spectrum follows after completion.

Moreover

$$\mathbf{F}_p \otimes R_p \simeq \mathbf{F}_p[x_0, x_1, \dots]$$

is discrete. Thus R_p is a Moore $\mathbb{E}_\infty$ -ring and

$$\pi_0 R_p \cong \mathbf{Z}_p[\widehat{x_0, x_1, \dots}]_p.$$

The δ_p -structure on $\pi_0 U$ is homogeneous and satisfies $\delta_p(x_i) = x_{i+1}$ by [Theorem 6.1](#); completion is compatible with the CNY construction. The recognition statement of [Theorem 3.2](#) therefore identifies the completed δ_p -ring with A . $\square$

The preceding theorem is the local form of the desired result. We next assemble compatible completions at all primes. Only the p -component is required to carry the prescribed free δ_p -structure.

Theorem 6.3 (Integral spherical lift). *For every prime p there is a Moore $\mathbb{E}_\infty$ -ring R and a ring isomorphism*

$$\theta : \mathbf{Z}\{x\} = \mathbf{Z}[x_0, x_1, \dots] \xrightarrow{\sim} \pi_0 R$$

such that, after p -completion, θ is an isomorphism of δ_p -rings

$$\widehat{\mathbf{Z}\{x\}}_p \xrightarrow{\sim} (\pi_0 R)_p^\wedge, \quad \delta_p(x_i) = x_{i+1}.$$

No condition is imposed on the δ_ℓ -structures for primes $\ell \neq p$.

Proof. Let R_p be the p -complete lift of [Theorem 6.2](#). For every prime $\ell \neq p$, let

$$R_\ell := \mathbb{S}[\mathbf{N}^{(\mathbf{N})}]_\ell^\wedge,$$

the ℓ -complete spherical monoid ring on countably many generators. By [\[CNY26, Construction 2.3 and Lemma 2.40\]](#), it is a Moore $\mathbb{E}_\infty$ -ring and

$$\pi_0 R_\ell \cong \mathbf{Z}[x_0, x_1, \dots]_\ell^\wedge.$$

At the prime p we use the corresponding ring isomorphism from [Theorem 6.2](#). In every case the underlying spectrum is the completion of a countable wedge of spheres, indexed by monomials.

Put

$$\Pi := \prod_{\ell} R_\ell, \quad \Pi_{\mathbf{Q}} := \Pi \otimes \mathbb{S}_{\mathbf{Q}}, \quad F := H\mathbf{Q}[x_0, x_1, \dots].$$

The ring F is the free commutative $H\mathbf{Q}$ -algebra on countably many degree-zero generators; rationally, its underlying spectrum is discrete. Let

$$a : F \longrightarrow \Pi_{\mathbf{Q}}$$

be the $\mathbb{E}_\infty$ - $H\mathbf{Q}$ -map sending x_i to the image of the compatible element $x_i \in \pi_0 R_\ell$ in every factor, and define

$$R := F \times_{\Pi_{\mathbf{Q}}} \Pi$$

in $\mathrm{CAlg}(\mathrm{Sp})$.

We first calculate its homotopy groups. For $n \geq 1$, the stable homotopy group $\pi_n \mathbb{S}$ is finite; hence its ℓ -primary component is nonzero for only finitely many ℓ . It follows that $\pi_n \Pi$ has bounded exponent and therefore

$$\pi_n \Pi_{\mathbf{Q}} = 0 \quad (n \neq 0).$$

The Mayer–Vietoris sequence for the pullback thus gives

$$\pi_n R \cong \pi_n \Pi \quad (n \geq 1),$$

and an exact sequence

$$0 \longrightarrow \pi_0 R \longrightarrow \mathbf{Q}[x_0, x_1, \dots] \oplus \prod_{\ell} \pi_0 R_{\ell} \xrightarrow{\alpha} \left(\prod_{\ell} \pi_0 R_{\ell} \right) \otimes \mathbf{Q} \longrightarrow \pi_{-1} R \longrightarrow 0,$$

where $\alpha(q, s) = a(q) - s \otimes 1$.

The map α is surjective. Indeed, write an element of the target as $(r_{\ell})_{\ell}/N$. For each prime $\ell \mid N$, approximate the ℓ -adic series r_{ℓ} by a polynomial $P_{\ell} \in \mathbf{Z}[x_0, x_1, \dots]$ modulo $\ell^{v_{\ell}(N)}$. The Chinese remainder theorem, applied coefficientwise to the finitely many primes dividing N , gives a polynomial Q satisfying all these congruences. Then

$$s_{\ell} := \frac{Q - r_{\ell}}{N} \in \pi_0 R_{\ell}$$

for every ℓ , and $\alpha(Q/N, s) = (r_{\ell})_{\ell}/N$. Hence $\pi_{-1} R = 0$.

Likewise, an element $q \in \mathbf{Q}[x_0, x_1, \dots]$ occurs in $\ker \alpha$ precisely when it is integral in every ℓ -adic completion. Writing $q = Q/N$, this says coefficientwise that every coefficient of Q is divisible by $\ell^{v_{\ell}(N)}$ for every ℓ , hence $q \in \mathbf{Z}[x_0, x_1, \dots]$. Therefore

$$\pi_0 R \cong \mathbf{Z}[x_0, x_1, \dots],$$

with the displayed generators.

It remains to check the Moore condition. Smashing the pullback square with $H\mathbf{Z}$ is exact. Since the rings R_{ℓ} are uniformly connective and $H\mathbf{Z}$ is bounded below of finite type, the natural map

$$\left(\prod_{\ell} R_{\ell} \right) \otimes H\mathbf{Z} \longrightarrow \prod_{\ell} (R_{\ell} \otimes H\mathbf{Z})$$

is an equivalence: in any fixed degree one may replace $H\mathbf{Z}$ by a finite skeleton. Each $R_{\ell} \otimes H\mathbf{Z}$ is discrete. Thus $R \otimes H\mathbf{Z}$ is the pullback of discrete Eilenberg–MacLane rings corresponding to the arithmetic square above. Since α is surjective, this pullback is discrete with

$$\pi_0(R \otimes H\mathbf{Z}) \cong \mathbf{Z}[x_0, x_1, \dots].$$

Hence R is a Moore $\mathbb{E}_{\infty}$ -ring.

Finally, the projection $R \rightarrow R_p$ is an $\mathbb{E}_{\infty}$ -map sending every x_i to the class with the same name. The Tate–Frobenius construction is natural on Moore $\mathbb{E}_{\infty}$ -rings [CNY26, Proposition 2.41]; therefore the induced map on p -completed π_0 is a map of δ_p -rings. It is an isomorphism of the underlying rings and, by Theorem 6.2, carries $\delta_p(x_i)$ to x_{i+1} . This proves the assertion. $\square$

Theorem 6.4 (Answer to Carmeli–Nikolaus–Yuan). *For every prime p , the free δ_p -ring on one generator admits a spherical lift. More precisely:*

- (i) *its p -completion admits a p -complete Moore $\mathbb{E}_{\infty}$ -lift whose Tate–Frobenius δ_p -structure is the free one;*
- (ii) *there is an integral Moore $\mathbb{E}_{\infty}$ -ring R with $\pi_0 R \cong \mathbf{Z}\{x\}$ for which the induced δ_p -structure after p -completion is the free one.*

Thus Question 3 of Carmeli–Nikolaus–Yuan [CNY26, Question 3] has an affirmative answer for every prime.

Proof. Part (i) is Theorem 6.2, and part (ii) is Theorem 6.3. $\square$

Remark 6.5. The construction is stronger than a bare existence result: the p -complete lift is obtained from the connected graded algebra of [Theorem 6.1](#), in which $|x_i| = p^i$ and $\delta_p(x_i) = x_{i+1}$. We do not assert a universal mapping property for this lift. Such a statement would require additional control of mapping spaces beyond the existence and Tate–Frobenius calculations carried out here.

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